Computers and Chemical Engineering 155 (2021) 107513

Contents lists available at ScienceDirect

Computers and Chemical Engineering

journal homepage: www.elsevier.com/locate/compchemeng

Crude oil price prediction: A comparison between AdaBoost-LSTM and AdaBoost-GRU for improving forecasting performance

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ARTICLE INFO

Article history:
Received 15 March 2021
Revised 1 August 2021
Accepted 28 August 2021
Available online 30 August 2021

Keywords:
Crude Oil Price Forecast
AdaBoost algorithm
Long short-term memory
Gated recurrent unit
Forecasting

ABSTRACT

Crude oil plays an important role in the world economy and contributes to more than one third of energy consumption worldwide. The better forecasting of its fluctuating price is crucial for policymaking. Although various methods had been previously applied for forecasting crude oil price which including both statistical models and Artificial Neural Network models but we find that there is no single paper that compares the AdaBoost-LSTM and AdaBoost-GRU models for improving forecasting performance. We proposed AdaBoost-GRU in which the GRU model was built, put inside sklearn wrapped and finally boost by AdaBoost Regressor. The predictive power of the proposed model was compared with AdaBoost-LSTM using daily Crude oil prices from October 23, 2009, to June 23, 2021, and single LSTM and single GRU were used as the benchmarking models. Forecasting performances were measured using five different metrics namely; the mean absolute error (MAE), the root mean squared error (RMSE), the Scatter Index (SI), the mean absolute percentage error (MAPE), and the weighted mean absolute percentage error (WMAPE). We have demonstrated that the AdaBoost-LSTM and the AdaBoost-GRU models outperform the benchmarking models as expected, and the empirical results show that the AdaBoost-GRU is superior to all models studied in this research.

1. Introduction

In the development of the World economy in general and the Nigerian economy in particular, crude oil plays an important role that requires more intention to predict the extent of its price volatility and irregularity for policymaking. Crude oil is an important commodity for economic development in both industrialized and developing countries. Fluctuation in oil price affects many goods and services that directly affect the economy, and pose a macroeconomic risk to both importing and exporting countries (Beckers, 2015). There is a lot of literature on forecasting crude oil prices using different methods with different data lengths, such as daily, weekly or monthly data in time series forecasting. A good forecasting method with a good data length, that is, a daily database that can predict the prices of crude oil on the daily basis should be adopted, given its cascading effect on the global economy. For instance, a monthly basis of 300 observations corresponding to data from 25 years may not be appropriate for the current economic situation (Elshendy et al., 2018)

Many researchers have implemented common traditional statistical and econometric techniques in forecasting international crude oil prices. Popular conventional statistical and econometric techniques include the autoregressive integrated moving average (ARIMA) model (Xiang, and Zhuang, 2013; Zhao and Wang, 2014), Random walk (RW) (Yu et al., 2017), generalized autoregressive conditional heteroskedasticity (GARCH) (Ahmed and Shabri, 2014); Wei et al., 2010), vector autoregressive (VAR) models (Ramyar and Kianfar, 2019) and error correction model (ECM) (Brigida and Mathew, 2014). Econometric models such as the ARIMA model, which assumed a linear relationship between independent and dependent variables (Kane et al., 2014) have been challenged for not being able to understand the complexity and nonlinearity of the financial time series that cause prediction errors (Sun et al., 2018). Thus, nonlinear and more complex artificial intelligence methods have recently been applied to cater to these obstacles. These methods include artificial neural networks (ANNs) (e.g. Abdullah and Zeng, 2010; Haidar et al., 2008; Ramyar and Kianfar, 2019; Rast, 2001); support vector machine (SVM) (e.g., Ahmed and Shabri, 2014; Chiroma et al., 2014; Xie et al., 2006; Yu et al., 2017); and deep learning method (e.g., Bristone et al.,

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https://doi.org/10.1016/j.compchemeng.2021.107513
0098-1354/

2019; Zhao et al., 2017). Zhao et al. (2017) forecast WTI oil prices using stacked denoising autoencoders (SDAE) and bagging. The empirical results from all these models reflected their superiority over econometric and statistical models.

ANNs are commonly used methods for a single model and a hybrid forecasting model. Although the combined forecasting models sometimes do not outperform the best single forecasting models, nearly all the combined forecasting models have better forecasting accuracy than the average or worst single forecasting models because they could considerably reduce the risk of forecasting failure in many real-world situations (Wong et al., 2007). Chang et al. (2020) exploit a stacking-based ensemble learning scheme with a Pearson correlation coefficient to compute the correlation between different machine learning models and construct machine learning-based frameworks for Spark + Hadoop and TensorFlow deep learning framework to forecast air pollution. To ensure the success of a combination of models, Sun and Li (2008) state that the individual model must have the principle of both diversity and performance. Wang and Ma (2012) propose an RSB-SVM model based on two ensemble strategies, i.e., bagging and random subspace, and uses Support Vector Machine (SVM) as a base learner for the assessment of enterprise credit risk and they show that RSB-SVM can be used as an alternative method for enterprise credit risk assessment. Yu et al. (2016) proposed a hybrid grid-GA-based LSSVR learning model to predict the crude oil spot prices of the WTI and the Brent markets and demonstrate that the proposed method is one of the effective forecasting tools for the crude oil price.

With the exception of Recurrent Neural Networks (RNNs), most neural networks (NNs) do not have a memory to process individually distinct input shown to them, and no state is maintained between inputs (Chollet and Allaire, 2018). A three-layer Neural Network (NN) consists of an input layer, a hidden layer, and finally an output layer. RNNs are powerful and robust types of NNs. They possess an internal memory that can remember important information about the input they received to adequately forecasting what is happen next. RNNs adopt a structure similar to other types of neural networks with modifications to take into account the hidden layers of the previous time steps in the current time step. The two variants of RNN are Long Short Term Memory (LSTM) and Gated Recurrent Unit (GRU). The data in these networks (that is RNN and its variants) flow in any direction within groups and across groups. One major issue with simple RNN is that it should be able to retain information about inputs seen many time-steps before but the problem of vanishing gradient makes such long-term dependencies difficult to learn. The LSTM layer developed by Hochreiter and Schmidhuber (1997) and GRU layers developed by Chung et al. (2015) in 2014 were designed to solve this problem. The LSTM model and the GRU model are widely used because they are both better than other methods and they can reasonably prompt peak flow periods (Apaydin et al., 2020).

In the past, many methods have been applied to predict crude oil prices but none of them implemented a hybrid approach comparing AdaBoost algorithm and Long Short-Term Memory (LSTM) network and AdaBoost algorithm and Gated Recurrent Unit (GRU) network. We aim at forecasting the crude oil price by comparing AdaBoost-LSTM with AdaBoost-GRU for improving forecasting accuracy. Sun et al. (2018) previously predicted financial time series comprised of two typical stock indexes and two main exchange rates and showed that the AdaBoost-LSTM ensemble-learning approach is superior to other single forecasting models and ensemble learning approaches used in the paper. Therefore, this paper is intending to compare the AdaBoost-LSTM model with the proposed AdaBoost-GRU model, and check their performances against the results of the two variants of RNN (that is, LSTM and GRU). The structure of this paper is as follows: after this introduction, section 2 presents the proposed methodology. Section 3 describes data collection and processing and evaluation metrics, and section 4 accommodates empirical findings on the crude oil prices while section 5 presents conclusions.

2. Methodology

2.1. Long Short-Term Memory (LSTM)

One of the variations of the RNN architecture is the Long Short-Term Memory (LSTM) explicitly designed to avoid problems with long-term dependency. Although the LSTM has a similar structure to RNN but the vanilla LSTM has three gates (that is, input, forget, and output), block input, a single cell, an output activation function, and peephole connections (Greff et al., 2016). The LSTM was the first recurrent network architecture to overcome the vanishing and exploding gradient problem. The forget gate in LSTM decides the information to pass through or the information to throw away from the cell state, the input gate determines what new information should be stored in the cell state while the output gate regulates what each cell produces. Moreover, this will depend on the state of the cell, the filtered, and newly added data. Fig.1 shows a diagram of the Long Short-Term Memory Cell. Each gate corresponds to a weight matrix and can be learned when training the network

A schematic of the vanilla LSTM has three gates (that is, input, forget and output), block input, a single cell, an output activation function, and peephole connections

$$i_t(input\ gate) = \sigma(W_i h_{t-1} + v_i x_t + b_i) \tag{1}$$

Eq. (2) below shows the expected value of the current state and the output from the input gate

$$\hat{C}_t = tanh(W_c h_{t-1} + v_c x_t + \ b_c) \tag{2}$$

$$f_t(forget\ gate) = \sigma(W_f h_{t-1} + v_f x_t + b_f) \tag{3}$$

$$O_t(output\ gate) = \sigma(W_o h_{t-1} + v_o x_t + b_o) \tag{4}$$

In the above equations, W_i, W_c, W_f and W_o represent the weight matrices of the input while v_i, v_c, v_f, and v_o are the weight matrices of recurrent connections. The subscript i, c, f, and o stand for the input gate, the memory cell, the forget gate, and the output gate, respectively. And,$b_i,\ b_c,\ b_f,\ and\ b_o$ are the respective biases of each gate while x_t stands for the input to the memory cell.

The Eqs. (1), (3), and (4) represent the input gate, the forget gate, and output gate, respectively. Eq. (2) is used to update the forget state which is given in Eq. (5) below.

$$C_t = f_t.C_{t-1} + i_t.\hat{C}_t \tag{5}$$

Finally, Eq. (6) below represents the hidden state computed by filtering the cell c_t with the output gate after the application of the hyperbolic tangent (tanh) activation function.

$$h_t = O_t\ * \tanh(C_t) \tag{6}$$

2.2. Gated recurrent units (GRUs)

GRU layers work by using the same principle as LSTM, but it merges the input gate and the forget gate, also merges the cell state and the hidden state. Therefore, the GRU can be described as a simpler gating mechanism that requires fewer parameters than the LSTM (Chung et al., 2015). This makes the GRU cheaper to run. The GRU layer is better at remembering the recent past information than the distant past to perform the present task, and the more recent data points are naturally more predictive than older

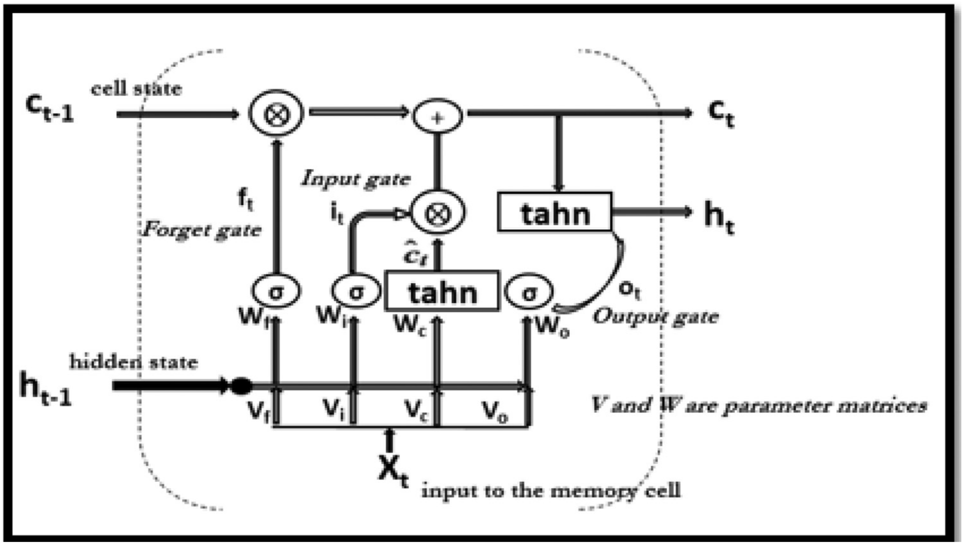


Fig. 1. the LSTM Cell Diagram (Source: Authors).

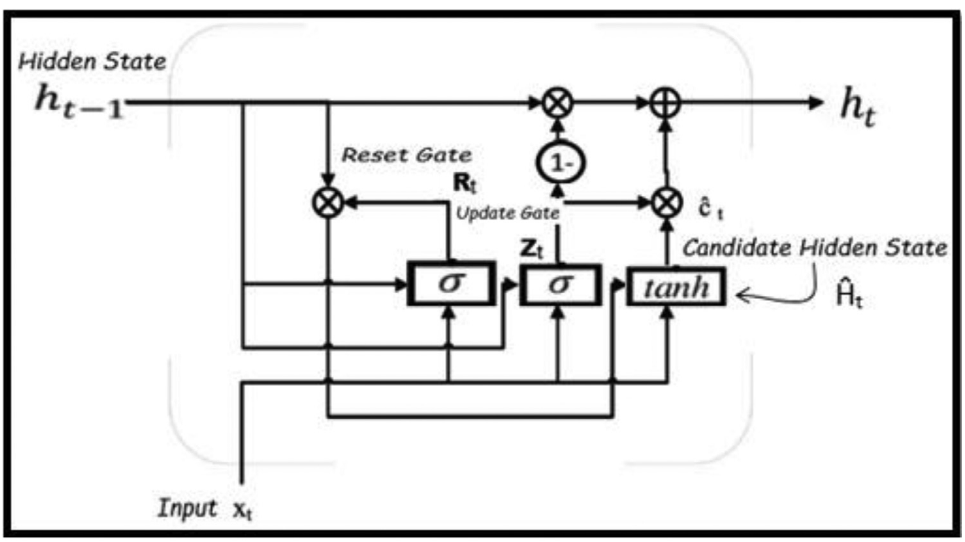


Fig. 2. the GRU Network Diagram (Source: Authors).

data points (Chollet and Allaire, 2018). The GRU consists of a reset gate and an update gate as shown in Fig. 2. The reset gate mats the new input to the previous memory. That is, it decides how much previous information should be forgotten. The update gate, on the other hand, defines the amount of the previous memory to store. It works similarly to the forget gate and the input gate in LSTM.

$$Z_t(Update\ Gate) = \sigma(W_Z\ .\ [h_{t-1}, x_t] + b_Z) \tag{7}$$

$$R_t(Reset\ \ Gate) = \sigma(W_R\ .\ [h_{t-1}, x_t] + b_R) \tag{8}$$

$$\hat{H}_t = \tanh(W.[R_t * +h_{t-1}, x_t] + b_h) \tag{9}$$

$$h_t = (1 - Z_t) * h_{t-1} + Z_t * \hat{H}_t) \tag{10}$$

2.3. The difference between the LSTM and the GRU

There are differences between the GRU and the LSTM and these are that, firstly, the LSTM has three gates consist of input, output, and forget gate but the GRU has two gates, which are reset and update. Secondly, the GRU possesses the same internal memory as the exposed hidden state. Thirdly, the GRU exposes the entire memory and hidden layers whereas the LSTM does not. Fourthly, the GRU is well applying if the dataset is small while LSTM relatively does well for the larger dataset. Finally, there is no application of second nonlinearity when computing the output in the GRU.

2.4. AdaBoost algorithm

The AdaBoost algorithm is a short form of Adaptive Boosting introduced by Freund and Schapire (1997). This method reduces bias by combining different machine-learning techniques into a single forecasting model. In boosting, the observations are weighted so that some of them participate repeatedly in the new combinations. Each classifier is also trained on data considering the success of the previous classifiers. After each training step, the weights are redistributed. Misclassified data is weighted more to highlight the most difficult cases. This way, subsequent learners will focus on them during their training. Table 1 shows the application of the AdaBoost algorithm.

Table 1
Summary of the usefulness of AdaBoost algorithm.

Performance Boosting	Achieving Goal	Applying area	Function
By giving higher weightage for misclassified classes samples	coming up with a strong classifier which increases the predictive force with surprising accuracy	Gradient descent	Using of a base weak learner to form a strong learner by weighted addition of the weak learners

2.4.1. Approach to AdaBoost-LSTM and AdaBoost-GRU

Having prepared the data, we start by inputting the sequence of, say m samples $(x_1, y_1) \ldots (x_m, y_m)$, where output "y'' is such that (y ∊ R) Moreover, we create the model for a base weak learner, that is, LSTM and GRU. We, then, initializing the sampling weights (W_i) of the training sample set. Then the created models will be put in the scikit-learn wrapper and finally boosted. The steps are as follow:

Initialize sampling weights W_i of the training sample. The weights are set equally, where:

$$W_i = \frac{1}{N}, i = 1, 2, 3, \ldots, N \tag{11}$$

N is the number of observations for both LSTM and GRU,

Training of LSTM(GRU) by training samples and compute the error following the second step of AdaBoost Algorithm (Dangeti et al., 2018.). For $m = 1$ to M

Fit a classifier $H_m(x)$ to the training data by minimizing weighted error functionF_m:

Where F_m is

$$F_m = \sum_{i=1}^{N} W_i^{\,m} \; I[y_i \neq H_m(x_i)] \tag{12}$$

$$Compute\, err_m = \frac{F_m}{\sum_{i=1}^{N} W_i^{\,m}} \tag{13}$$

$$Evaluate\, \alpha_m = log\left(\frac{1 - err_m}{err_m}\right) \tag{14}$$

Then update the sampling weights by setting.

$$W_i^{(m+1)} = W_i^m exp[\alpha_m * I(y_i \neq H_m(x_i))] \tag{15}$$

Iterate steps 2 and 3 until all the predictors for both networks are obtained

Then finally, output the predicting result by combining the predicting results of all the predictors with ensemble weights as the final step of the AdaBoost Algorithm

$$H_m(x) = argmax\left[\sum_{m=1}^{M} \alpha_m H_m(x)\right] \tag{16}$$

3. Data Collection and Processing

3.1. Data collection

For better performance, ANN algorithms require more data points. For this reason, the intraday or daily data where a large number of observations can be obtained in a shorter period is appropriated. Because the economic conditions change over time, training the network with old information could result in a poor model generalization that might not be relevant to the present economic situation (Elshendy et al., 2018). In this study, the daily data for the last eleven years and eight months, that is, from October 23, 2009, to June 23, 2021, predicting the export crude oil price was used. This is graphically shown in Fig. 3, and Table 2

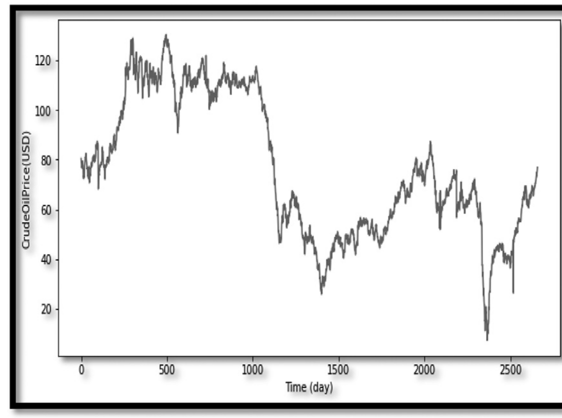


Fig. 3. Daily Export Crude Oil Spot Price (USD$).

Table 2
The Descriptive Statistics of Daily Export Crude Oil Spot Prices (USD$).

COUNT	MEAN	STD	MIN	25%	50%	75%	MAX
2660.0	76.38	28.09	7.15	52.95	70.04	108.54	130.43

shows the descriptive statistics of this research data. The data were obtained from the Central Bank of Nigeria Statistical Bulletin database and it can be found on the following website: https://www.cbn.gov.ng/rates/DailyCrude.asp. From Table 2, we can see that the minimum price for those periods is 7.15USD and the maximum export price is 130.43USD.

3.2. Data processing

Preprocessing of the data in ANN is essential for improved forecasting accuracy (Ahmed et al., 2010; McAdam and McNelis, 2005; Zhang et al., 1998) and has the advantage of reducing the risk of computational problems to make the training process more efficient (Zhang et al., 1998). The first step of processing consists of normalizing the data using a min-max scaler. We considered the [−1, 1] to perform our analysis. In the output layer of the AdaBoost-LSTM and the AdaBoost-GRU, it is necessary to set the scale range from −1 to +1 for the input and output variables to match the scale of the activation function (that is, hyperbolic tangent)

$$X_{norm}[min - \max Scaler] = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{17}$$

Using a forward-moving window of size 5, we used the first 5 data points as input (predictor, that is, x1 to x5) to predict the sixth data point (target, that is, y1) and we used the window between second to sixth data points as input (that is, x2 to x6) to predict the seventh data point and so on. To fix the size of moving window to five, we use the pandas shift function to shift the entire column by the number we specified. The data consists of 2655 rows with six columns. The effect of the training data and

Table 3
Train and test dataset of Daily Crude Oil Spot Prices showing the dimension of each as well as predictors and targets' column.

Splitting Dataset	Dimension of data	Predictors' Column	Targets' Column
Train Data	1991, 6	1~5th	6th
Test Data	664, 6	1~5th	6th

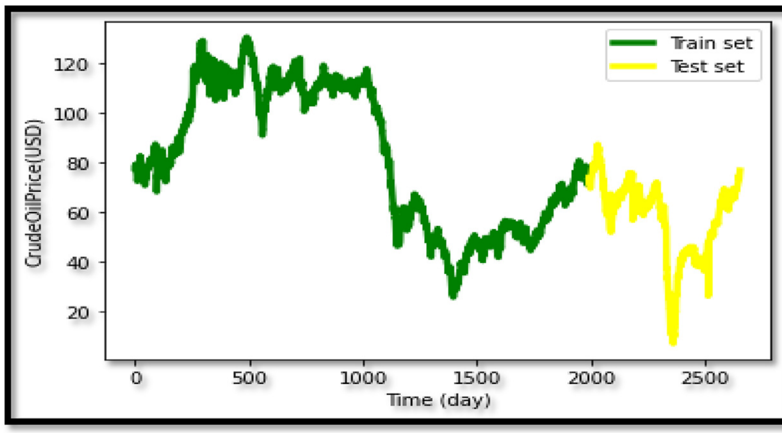


Fig. 4. Train and Test dataset of Daily Crude Oil Spot Prices where the green line is the train set represent 1991 of total data and the yellow line is the test set.

the testing data were investigated by splitting them into 75% and 25%, respectively, as illustrated in Table 3 and Fig. 4. Thus, fixing the window of size 5 constitutes the first 1991 observations that account for 75% of the total data as the training set and the remaining 664 observations are for the test set.

Assuming that the dataset of observation denotes by K, so that $K = (x_i, y_i)_{i=1}^N \cdot (x_i)_{i=1}^N$ are the input data to the model while $(y_i)_{i=1}^N$ are the labels. Dimensionall y, $x_i \in \mathbb{R}^d$ where, 'd' is the number of variables. In the case of our work, 'd' is 5, and $y_i \in \mathbb{R}$

Then, $|K| = N$ where N is the total number of observations

The dataset splits into the training set and the test set. If the training set is denoted by P and the test set by Q, then,

$K = P \cup Q$, where P and Q are disjoint (that is, $P \cap Q = \emptyset$)

Furthermore, if the sample size of the training set is 'a' (that is $|P| = a$) and that of test set is 'b' (that is, $|Q| = b$), then $b = N$-a

Therefore, from the dataset,

The training set: $P = (x_i, y_i)_{i=1}^b$ while the test set: $Q = (x_i, y_i)_{i=1}^b$ Note that $a + b = N$

After renaming the columns' names for convenient purposes, we use column 'y', that is, the sixth day for the first row as the target variable (output) while the first five columns named; 'a', 'b', 'c', 'd', and 'e' are for the predictors' variables (input).

The input data and the target data of Keras LSTM, GRU, AdaBoost-LSTM, and AdaBoost-GRU are expected to have a fixed three-dimensional shape. These are the number of observations to be given to the model, the length of the hidden state, and the number of predictors. When the models were built for LSTM and GRU, there were two hidden layers that included 256 neurons, one neuron in the output layer, and the dropout functions were set to 0.5. The cost functions were calculated as the mean squared error while the optimizer was Adaptive Moment Estimation (Adam). Furthermore, when fitting the models, epochs were set to 100 with batch sizes of 10. Since the order of data is important in our analysis, random (that is, shuffle) was set to 'false', and the verbose was set to one. When the LSTM model and the GRU model were created, the AdaBoost algorithm was applied by putting them the inside sklearn wrapper and finally boosted. Fig. 5 above shows the stages of data processing, which includes how the AdaBoost algorithm is applied. The AdaBoost implementation on any model involves the wrapper of the defined model and boosts it with either AdaBoost classifier or AdaBoost Regressor as the case may be.

3.3. Evaluation metrics

There are several methods for assessing predictive performance, with different focus and characteristics. To evaluate the forecasting performance of the proposed AdaBoost-GRU compared to AdaBoost-LSTM and other benchmarking models, five different statistics, namely, the mean absolute error (MAE), the root mean squared error (RMSE), the scatter index (SI) the mean absolute percentage error (MAPE), and the weighted mean absolute percentage error (WMAPE) were employed.

Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are the most widely used forecasting metrics by statisticians. While MAE and RMSE are easy to compute, the values in RMSE are difficult to interpret compared to MAE. Both of these metrics are scale-dependent errors meaning that we cannot use these measures to compare the results of two different time-series forecasts with different units Chatfield, 1988). Since the RMSE values do not tell us whether the estimations are good or not, the term called scatter index (SI) that is defined to determine whether the RMSE is good or not is also used. The SI, which is a normalized RMSE is calculated by dividing the RMSE by the data average and the estimations are acceptable if the value of SI is less than one. The values obtained here are converted into percentages. Eqs. (18)–((20) are the mathematical representation of the MAE and RMSE

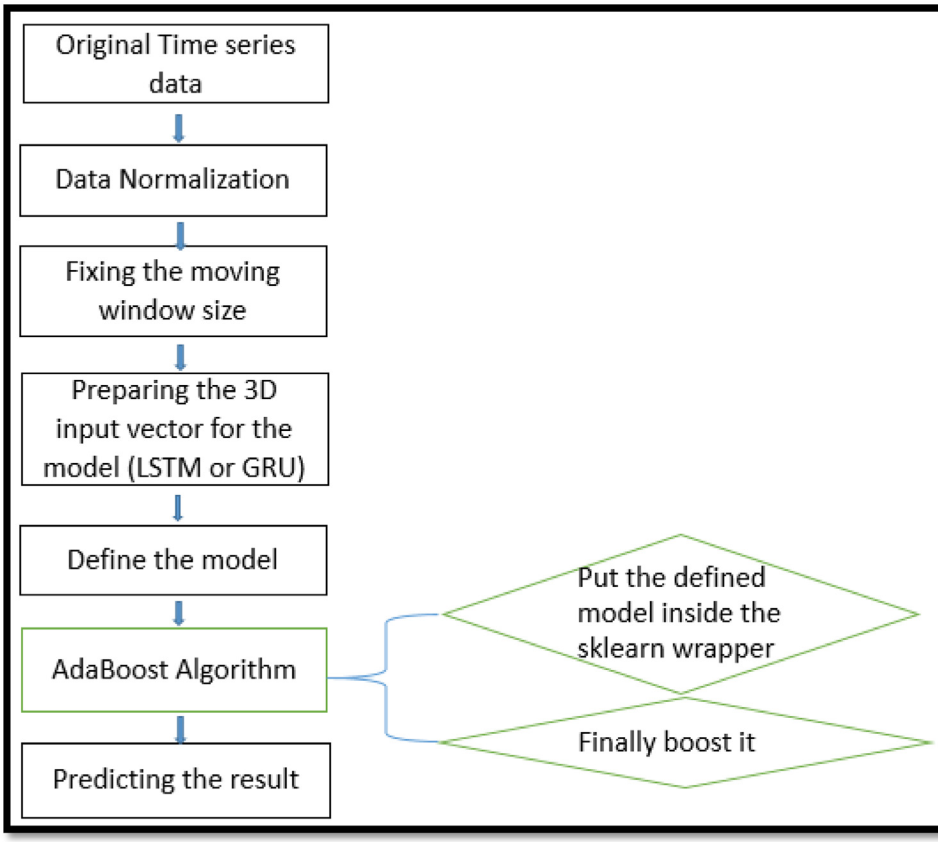


Fig. 5. The flowchart of data processing stages including how the AdaBoost algorithm is applied.

metrics and SI, respectively.

$$MAE = \frac{\sum_{i=1}^{N} |F_i - A_i|}{N} \tag{18}$$

$$RMSE = \frac{\sqrt{\sum_{i=1}^{N} (A_i - F_i)^2}}{N} \tag{19}$$

$$SI = (\frac{RMSE}{dm}) \tag{20}$$

The evaluation metrics are a very important aspect of the papers and should be taken into account at the core of any paper, as they actually tell us the degree of effectiveness of the methods employed in the paper. For this reason, the paper also used Mean Absolute Percentage Error (MAPE) and Weighted Mean Absolute Percentage Error (WMAPE), which are scale-independent. Thus, they are safe to use for performances comparison of time series forecast values with different units. However, MAPE is not an appropriate method where there is a mix of fast and slow-moving changing of values as it does not understand this. Therefore, the Weighted Mean Absolute Percentage Error (WMAPE), which is a variant of MAPE to overcome the infinite error problems of MAPE, was applied. Eqs. (21) and (22) are the mathematical expression of the MAPE and WMAPE metrics, respectively.

$$MAPE = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{A_i - F_i}{A_i} \right| \tag{21}$$

$$WMPE = \frac{\sum_{i=1}^{N} |A_i - F_i|}{\sum_{i=1}^{N} |A_i|} \tag{22}$$

In the above metric equations, N is the number of data points, F_i is the predicted value, A_i is the actual observation and 'dm' is the data mean (or average).

4. Empirical result and discussion

Four forecasting models including two comparative analysis models (AdaBoost-LSTM and AdaBoost-GRU) and two benchmarked models (single LSTM and single GRU) were employed for the crude oil prices data to test the forecasting measure. The forecasting power of those models varies in performance with some hierarchical structures. We established for five different metrics used in the paper that the AdaBoost-GRU model outperforms all other models, making it a more realistic and powerful tool for forecasting complex time series problems.

To prevent overfitting of the test set, we started by training all networks for 100 epochs and dropout techniques were applied to increase efficiency. We used Adam's optimization approach and all the networks were trained with Keras in Python programming using google colaboratory application. The LSTM and the GRU neural networks permit the examination of the results of forecasting accuracy at the current time step on the data from the past time steps. An overview of the outcomes of the AdaBoost-LSTM and the AdaBoost-GRU models are compared with the single LSTM and GRU models serving as the benchmarks. Fig. 6 depicted the individual graphs of the predicted values and the actual values for each model. For instance, Fig. 6(b)–(e) provide visualization of the predicted trends for all the models starting from a single LSTM, a single GRU, AdaBoost-LSTM and AdaBoost-GRU, respectively while Fig. 6(a) shows trends of the original data. The trends, as depicted in Fig. 6(b)–(e), are similar to Fig. 6(a), which means they each have good prediction accuracy for a given level with respect to their individual proximity to reality. The reason for such a motivated behavior is that recurrent neural network variants are mainly good at forecasting smooth regression trends to produce predictive results from sequential data. Taking a close look, we can see that

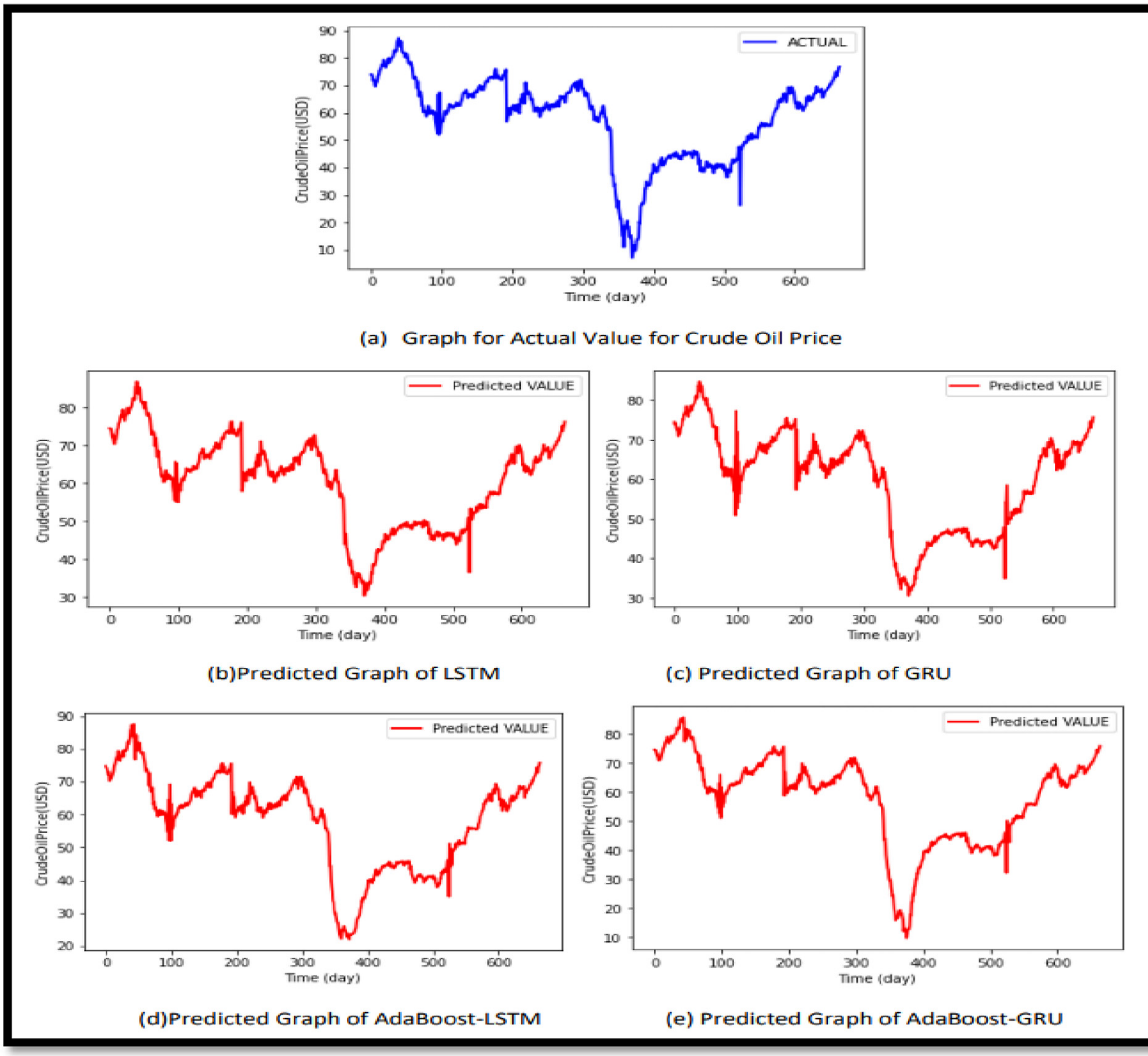


Fig. 6. The trends graph for (a) the Actual value of Crude oil price (b) the predicted value of LSTM, (c) the predicted value of the GRU, (d) the predicted value of the AdaBoost-LSTM, and (e) the predicted value of the AdaBoost-GRU.

Table 4
Metrics comparison for different forecasting Model approaches.

METRICS	LSTM	GRU	AdaBoost-LSTM	AdaBoost-GRU
MAE	3.2925	3.0372	1.6374	1.4164
RMSE	5.5120	5.2622	3.0161	2.4602
SI	0.722	0.0689	0.0395	0.0322
MAPE	0.1162	0.1071	0.0540	0.0354
WMAPE	0.0573	0.0529	0.0285	0.0247

they are slightly different from each other and that from the actual values graph. The slight differences can also be inferred from the values displayed on the y-axes, which represent the range of crude oil prices values included in each graph

The simulation outcomes for these models comparing the actual and the forecast values are shown in Fig. 7, where the blue lines represent the actual values and the red lines represent predicted values. The observations in Fig. 7 below show that the trends examine for both single LSTM and single GRU neural networks can also be observed with AdaBoost-LSTM and AdaBoost-GRU. It is also obvious that AdaBoost-GRU, as depicted in Fig. 7(d), has the greatest predictive power compared to the others. Both the AdaBoost-LSTM and the AdaBoost-GRU ensemble learning approach performed better than the single models as expected.

Table 4 show the comparison results of the five evaluation metrics, MAE, RMSE, SI, MAPE, and WMAPE, while the predictive performance of the AdaBoost-GRU approach outperforming all other models in this study. That is, the important outcome of the predictions measured by all of these metrics is that the AdaBoost-GRU model with the lowest yellow bars in all groups outperform all other models, as shown in Fig. 8. This is also numerically shown under the histogram. In summary, the AdaBoost-GRU model is seconds to none in terms of performance while the AdaBoost-LSTM also has better performance meaning that the hybrid methods give improved forecasting performance than the single models.

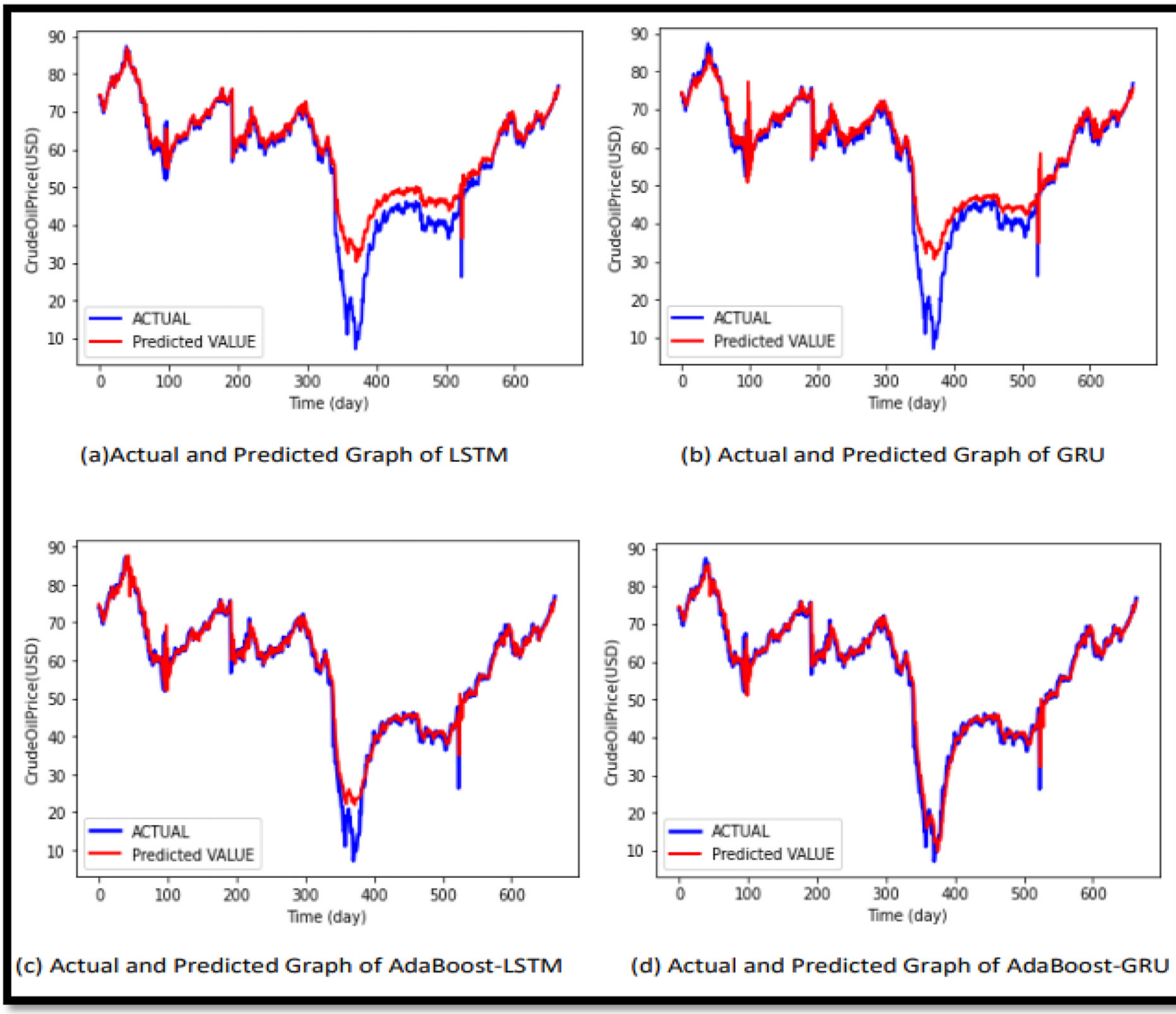
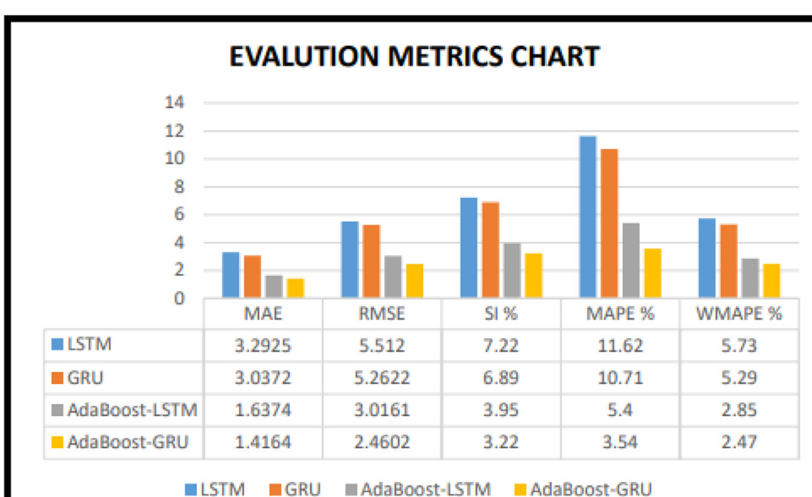


Fig. 7. The visual graphs of the trends for (a) the Actual and Predicted graph of LSTM (b) the Actual and Predicted graph of GRU, (c) the Actual and Predicted graph of AdaBoost-LSTM, (d) the Actual and Predicted graph of AdaBoost-GRU.



	MAE	RMSE	SI %	MAPE %	WMAPE %
LSTM	3.2925	5.512	7.22	11.62	5.73
GRU	3.0372	5.2622	6.89	10.71	5.29
AdaBoost-LSTM	1.6374	3.0161	3.95	5.4	2.85
AdaBoost-GRU	1.4164	2.4602	3.22	3.54	2.47

Fig. 8. Evaluation metrics chart that shows both the histogram and the values of the metrics for all models and present SI, MAPE, and WMAPE in percentage.

5. Conclusion

Given that the deep learning networks have the advantage of selecting features from a set of data without depending on the foreknowledge of predictors, this paper shows the possibility of improving forecasting performance by comparing the AdaBoost-LSTM and the AdaBoost-GRU ensemble learning methods. We checked the predicting powers of both of them against the single models, that is, the LSTM model and the GRU model on crude oil prices from October 23, 2009, to July 23, 2021. We measure forecasting performance in terms of five evaluation metrics. These metrics include both scale-dependence (e.g., MAE and RMSE) and scale-independence (e.g., MAPE and WMAPE) metrics. The scatter index (SI) was also used to determine the goodness of the RMSE. The results obtained indicate that the performances of the models are consistent from the single models having lower performances compared to the hybrid methods that show improved results. While a single GRU is better than a single LSTM, in this paper, the AdaBoost-GRU ensemble model that robustly delivers the smallest errors is superior to all of them include AdaBoost-LSTM. Consequently, the AdaBoost-GRU ensemble-learning model of all other models is a promising approach to forecasting crude oil prices.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

None.

CRediT authorship contribution statement

Ganiyu Adewale Busari: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Dong Hoon Lim:** Conceptualization, Methodology, Investigation, Supervision, Validation.

Acknowledgement

The careful reading of the manuscript and the insightful suggestions by Mudasiru, A. Busari (Mr.) is gratefully acknowledged.

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